(just as relevant in 2025 as it was in the 1970’s) deals with the challenges of representing reality through data and begins with the following quote

> *Entities are a state of mind. No two people agree on what the real world view is.*

— Apollon Metaxides
as quoted by William Kent

The significance of this is that humans perceive and manage data in different ways, and a data model is really a person’s perception of ambiguous information, not objective reality itself.

If this starts to sound too philosophical, lets consider a practical example that you may experience in your daily life. If you have both a checking account and a credit card issued by your bank, your information is likely stored in (at least) two different systems. You may be identified in both of those systems with a common identifier (say a customer number), but often this is not the case. Your name is probably recorded in both systems, but sometimes spelt differently. In the card system my name could be Gregory, but in the other system it is Greg. In order to obtain a full picture of a customer (the so called customer 360 view), my bank must link all the different digital representations together. The same is true across most industries today—think of your own industry or personal life and all the disconnected digital representations that exist.

The examples we have looked at so far of contextualizing data by linking data and entities together have been easy to do because of the small scale and the fact that I know the relationships. But what happens when you have hundreds of thousands of records spread across multiple systems where you do not know the relationships.

CONTEXTUALIZATION THROUGH DATA MATCHING

In practical terms linking related facts together means matching data stored in different systems and identified differently. This often entails matching different identifiers that are very similar but not identical. Think of the street address '51 Stratford Drive'—when written as '51 Stratford Dr.' a human can